

Benjamin D. Umans

Education

- November 2019 **Ph.D., Biological and Biomedical Sciences.**
Harvard University, Cambridge, MA
Dissertation: “Vagal afferent pathways of laryngeal and gastric sensation”
Thesis advisor: Dr. Stephen Liberles
- May 2014 **MPhil, Biological Science.**
University of Cambridge, Cambridge, UK
Thesis: “An investigation of physical domains in the *Drosophila* genome”
Thesis advisor: Dr. Rob White
- October 2012 **MSc with Distinction, Applied Statistics.**
University of Oxford, Oxford, UK
Thesis: “Modelling bacterial evolution from sequence data”
Thesis advisor: Dr. Daniel Wilson
- June 2010 **BA with Honors, Biology and Economics.**
The University of Chicago, Chicago, IL

Research Experience

- 2025–present **Research Scientist**, Department of Biology, Massachusetts Institute of Technology.
 - Identifying peripheral neural pathways and autonomic circuits that control respiratory physiology and tissue homeostasis
 - With the Biology of Adversity Project (Broad Institute), working to understand the effects of stress and adversity on endocrine and peripheral neural signaling
- 2020–2025 **Postdoctoral Scholar**, Section of Genetic Medicine, The University of Chicago.
Advisor: Dr. Yoav Gilad.
 - Using human and non-human primate stem cell platforms to study neuronal gene regulation in development and disease
 - Developed a brain organoid model to study cell type-specific effects of human gene regulatory variation on stress responses during brain development
- 9/2014–11/2019 **PhD Student**, Department of Cell Biology, Harvard Medical School, HHMI.
Advisor: Dr. Stephen Liberles.
 - Used mouse genetic tools, optogenetics, advanced imaging techniques, behavioral experiments, single-cell transcriptional profiling, and electrophysiology to study *in vivo* functions of molecularly defined sensory neurons and receptors.
 - Applied these tools to studies of autonomic sensory neurons in airway protection, gastrointestinal function, breathing, and neuroimmune regulation.

- 10/2012–8/2013 **MPhil Student**, University of Cambridge.
 Advisor: Dr. Rob White.
 ○ Investigated the relationship between chromatin state markers and genome architecture using HiC datasets
- 3/2012–9/2012 **MSc Student**, University of Oxford.
 Advisor: Dr. Daniel Wilson.
 ○ Developed models of evolution and natural selection to estimate species divergence times and substitution rates using whole-genome sequence data from human pathogens.
- 8/2011 **Visiting Researcher**, Texas Biomedical Research Institute.
 Advisors: Drs. Anthony Comuzzie and V. Saroja Voruganti.
 ○ Analyzed biomarker data and performed linkage analysis using data from two Native population cohort studies.

Awards and Fellowships

- 2022, 2024 **BSD CAP Award**, The University of Chicago.
 Competitive travel award for conference presentation.
- 2016-2019 **NRSA F31 Fellowship**, NIH, F31 HL132645.
- 2014-2015 **Cell and Developmental Biology Training Grant**, Harvard Medical School.
 Selected for inclusion on training grant 5T32GM007226 (Harper).
- 2011-2013 **Marshall Scholarship**.
 Selected as one of 33 American students funded to pursue graduate education in the U.K.
- 2010 **Phi Beta Kappa**, The University of Chicago.
- 2009-2010 **Student Marshal**, The University of Chicago.
 Highest undergraduate award.

Publications

Journal Articles

- [1] N. Almanzar, D. Yang, J. Xia, S. Udit, P. Joshi, S. Adhikari, D. A. Hoagland, S. T. Yeung, C. Khairallah, T. Huerta, A. Wallrapp, **B. D. Umans**, N. Sarden, O. Erdogan, N. Baalbaki, J. Hou, A. Beekmayer-Dhillon, J. Lee, K. A. Meerschaert, S. D. Liberles, R. A. Franklin, B. G. Yipp, K. M. Khanna, P. Baral, A. L. Haber, and I. M. Chiu. “Vagal TRPV1+ sensory neurons protect against influenza virus infection by regulating lung myeloid cell dynamics”. In: *Science Immunology* 10.110 (Aug. 2025), eads6243.
- [2] **B. D. Umans** and Y. Gilad. “Oxygen-induced stress reveals context-specific gene regulatory effects in human brain organoids”. en. In: *Genome Research* 35.8 (Aug. 2025), pp. 1689–1700.
- [3] **B. D. Umans**, A. Battle, and Y. Gilad. “Where Are the Disease-Associated eQTLs?” In: *Trends in Genetics* 37.2 (Feb. 2021), pp. 109–124.
- [4] S. L. Prescott*, **B. D. Umans***, E. K. Williams, R. D. Brust, and S. D. Liberles. “An Airway Protection Program Revealed by Sweeping Genetic Control of Vagal Afferents.” In: *Cell* 181.3 (Apr. 2020), 574–589.e14.
- [5] **B. D. Umans** and S. D. Liberles. “Neural Sensing of Organ Volume.” In: *Trends in Neurosciences* 41.12 (Dec. 2018), pp. 911–924.

- [6] A. I. Weaver, S. G. Murphy, **B. D. Umans**, S. Tallavajhala, I. Onyekwere, S. Wittels, J.-H. Shin, M. Van-Nieuwenhze, M. K. Waldor, and T. Dörr. “Genetic Determinants of Penicillin Tolerance in *Vibrio cholerae*.” In: *Antimicrobial Agents and Chemotherapy* 62.10 (Oct. 2018).
- [7] P. Baral, **B. D. Umans**, L. Li, A. Wallrapp, M. Bist, T. Kirschbaum, Y. Wei, Y. Zhou, V. K. Kuchroo, P. R. Burkett, B. G. Yipp, S. D. Liberles, and I. M. Chiu. “Nociceptor sensory neurons suppress neutrophil and $\gamma\delta$ T cell responses in bacterial lung infections and lethal pneumonia.” In: *Nature Medicine* 24.4 (May 2018), pp. 417–426.
- [8] S. El-Sharnouby, B. Fischer, J. P. Magbanua, **B. Umans**, R. Flower, S. W. Choo, S. Russell, and R. White. “Regions of very low H3K27me3 partition the *Drosophila* genome into topological domains.” In: *PLoS ONE* 12.3 (2017), e0172725.
- [9] T. Dörr, F. Delgado, **B. D. Umans**, M. A. Gerding, B. M. Davis, and M. K. Waldor. “A Transposon Screen Identifies Genetic Determinants of *Vibrio cholerae* Resistance to High-Molecular-Weight Antibiotics.” In: *Antimicrobial Agents and Chemotherapy* 60.8 (Aug. 2016), pp. 4757–4763.
- [10] R. L. Umans, **B. D. Umans**, H. Umans, and E. Elsinger. “Predictive MRI correlates of lesser metatarsophalangeal joint plantar plate tear.” In: *Skeletal Radiology* 45.7 (July 2016), pp. 969–975.
- [11] E. K. Williams, R. B. Chang, D. E. Storchlic, **B. D. Umans**, B. B. Lowell, and S. D. Liberles. “Sensory Neurons that Detect Stretch and Nutrients in the Digestive System.” In: *Cell* 166.1 (June 2016), pp. 209–221.
- [12] R. B. Chang, D. E. Storchlic, E. K. Williams, **B. D. Umans**, and S. D. Liberles. “Vagal Sensory Neuron Subtypes that Differentially Control Breathing.” In: *Cell* 161.3 (Apr. 2015), pp. 622–633.
- [13] S. O. E. Ebbesson, V. S. Voruganti, P. B. Higgins, R. R. Fabsitz, L. O. Ebbesson, S. Laston, W. S. Harris, J. Kennish, **B. D. Umans**, H. Wang, R. B. Devereux, P. M. Okin, N. J. Weissman, J. W. MacCluer, J. G. Umans, and B. V. Howard. “Fatty acids linked to cardiovascular mortality are associated with risk factors.” In: *International Journal of Circumpolar Health* 74.1 (Jan. 2015), p. 28055.

Presentations

Oral Presentations

- [1] **B. D. Umans**. “Genetic variation influencing brain oxygen stress responses contributes to disease risk.” Midwest Society for Developmental Biology. Madison, Aug. 2024.
- [2] **B. D. Umans**. “Sensory neuron populations contributing to airway protection.” Massachusetts Eye and Ear Infirmary Laryngology Research Rounds. Boston, Mar. 2019.
- [3] **B. D. Umans** and S. D. Liberles. “Deciphering sensory signals of the vagus nerve.” Dissecting microbiome-gut-brain circuits for microbial modulation of host cognition in response to diet and stress. Pasadena: MURI (Army Research Office), Feb. 2018.
- [4] **B. D. Umans**. “Mapping Chromatin Interactions in the *Drosophila* Genome.” Department of Physiology, Development and Neuroscience Graduate Symposium. Cambridge, 2013.

Oral Presentations presented by colleagues

- [1] R. L. Umans, **B. D. Umans**, H. Umans, and E. Elsinger. “Predictive MRI Correlates of Lesser Metatarsophalangeal Joint (MPJ) Plantar Plate (PP) Tear.” Society of Skeletal Radiology. Scottsdale, 2015.
- [2] H. Umans, W. Rennie, **B. D. Umans**, M. Shah, and H. Levy. “MRI Diagnosis of Glenoid Labral Tear Using the Biceps Labral Oblique (BLO) Sequence.” Society of Skeletal Radiology. San Diego, 2014.
- [3] R. L. Umans, **B. D. Umans**, H. Umans, and E. Elsinger. “Predictive MRI Correlates of Lesser Metatarsophalangeal Plantar Plate Tear.” Radiological Society of North America. Chicago, 2014.

- [4] R. L. Umans, **B. D. Umans**, H. Umans, and E. Elsinger. “Predictive MRI Correlates of Lesser Metatarsophalangeal Plantar Plate Tear.” International Skeletal Society. Edinburgh, 2014.

Poster Presentations

- [1] **B. D. Umans**, O. Allen, and Y. Gilad. “Genetic variation influencing brain oxygen stress responses contributes to disease”. American Society for Human Genetics. Denver, Nov. 2024.
- [2] **B. D. Umans**, O. Allen, and Y. Gilad. “Inter-individual genetic differences contribute to cell type-specific brain oxygen stress responses”. Society for Neuroscience. Chicago, Oct. 2024.
- [3] **B. D. Umans**, O. Allen, and Y. Gilad. “Mapping genetic contributions to individual- and cell type-specific brain oxidative stress responses”. American Society of Human Genetics. Washington, DC, Nov. 2023.
- [4] **B. D. Umans**, O. Allen, and Y. Gilad. “Mapping genetic contributions to individual- and cell type-specific brain oxidative stress responses”. Development & 3D Modeling of the Human Brain. Cold Spring Harbor Laboratory, Dec. 2022.
- [5] **B. D. Umans**, N. Joshi, E. K. Williams, S. L. Prescott, and S. D. Liberles. “Molecular diversity of vagal sensory neurons controlling airway protection.” Society for Neuroscience. Chicago, Oct. 2019.
- [6] **B. D. Umans**, S. L. Prescott, and S. D. Liberles. “Sensory innervation of the upper airway by Piezo2⁺ neurons.” Department of Cell Biology Retreat. Falmouth, Oct. 2018.
- [7] E. K. Williams, R. B. Chang, D. E. Storchlic, **B. D. Umans**, R. D. Brust, B. B. Lowell, and S. D. Liberles. “Sensory Mechanisms of Intestinal Vagal Afferents.” Giovanni Armenise-Harvard Foundation Symposium: From Molecular Mechanisms to Precision Medicine. Gubbio, June 2016.

Poster Presentations presented by colleagues

- [1] E. McIntire, K. Rhodes, K. Barr, M. DeMille, **B. Umans**, J. Burnett, Gonzales Natalia, and Y. Gilad. “Modeling cardiac cell developmental trajectories at high temporal resolution; (PB1534).” American Society of Human Genetics. Los Angeles, Oct. 2022.
- [2] V. S. Voruganti, **B. D. Umans**, K. Haack, S. Laston, R. R. Fabsitz, S. O. E. Ebbesson, J. W. MacCluer, B. V. Howard, J. G. Umans, A. G. Comuzzie, and S. A. Cole. “Genetic variants of SLC22A12 are associated in a population-specific manner with serum uric acid and CKD phenotypes.” American Society of Nephrology. Atlanta, 2013.
- [3] M. E. Bush, **B. D. Umans**, D. Choi, C. Voisine, D. Czyz, R. I. Morimoto, and R. F. Ismagilov. “A Microfluidic Platform for Local Thermal and Chemical Stimulation of *C. elegans*.” 13th Annual Midwest Stress Response and Molecular Chaperone Meeting. Evanston, 2008.

Teaching Experience

2020-present

Section of Genetic Medicine.

- Mentored two graduate students and one technician.

2014-2019

Department of Cell Biology.

- Mentored one graduate student and two technicians, as well as training new postdoctoral fellows in surgical and physiological measurement techniques.
- Trained researchers from the laboratories of Elaine Hsiao (UCLA) and Tom Rapoport (Harvard Medical School, HHMI) in experimental techniques, and provided ongoing technical guidance to researchers at UCLA and the Institut Pasteur.

2016-2017

Introduction to Histology for Graduate Students, Guest Lecturer.

Primary instructors: Drs. Gerald Greenhouse and Everett Anderson

- 2013-2015 **Pforzheimer House, Harvard College, Tutor in Statistics and Biology.**
 ○ Held weekly office hours and helped undergraduate students with course material and thesis research.
- 2014 **CURE Program, Dana-Farber Cancer Institute, Instructor.**
 ○ Taught students to read primary and secondary scientific literature related to clinical and basic studies in cancer as part of the Continuing Umbrella of Research Experience Program.

Service

- Reviewer. *Cell Reports, Annals of the American Thoracic Society*
- 2021-2025 Department of Human Genetics Postdoctoral Liaison. *The University of Chicago*
- 2014-2019 Cell Biology/BCMP Journal Club. *Harvard Medical School*
- 2015-2017 Admissions Committee. *Harvard Biological and Biomedical Sciences PhD Program*
- 10/2016 HMS Discovery Council. *Harvard Medical School*
- 2015 Conduct of Science Steering Committee. *Division of Medical Sciences, Harvard University*